VIS 2025



Beyond the Broadcast:
Enhancing VR Tennis Broadcasting through Embedded Visualizations and Camera Techniques

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Sports broadcasting Transformation

- How current technologies are reshaping the sports viewing experience?
- VR → Transportational Immersion
- AR → Distraction-Free Attention

COSM





https://www.cosm.com/

Challenges for Current VR Tennis broadcasting



180° 2D Monoscopic Video

- Fast rallies **overload attention**
- **Static** cameras limit perspective
- **Narrative guidance** is lacking



Lower Engagement



Reduced Comprehension

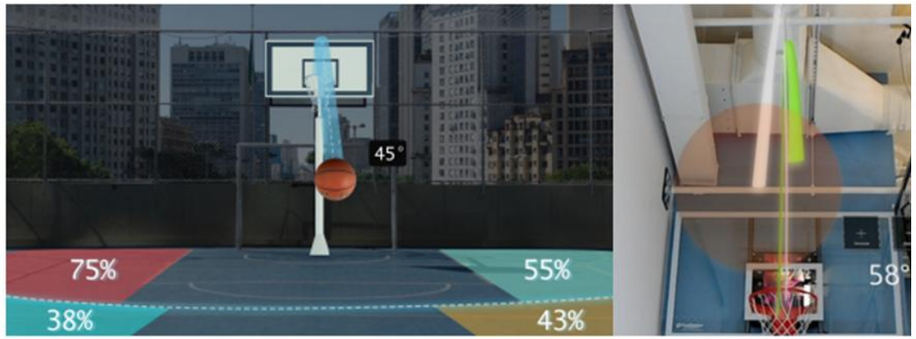
https://www.youtube.com/watch?v=Tk0Umjie9NA&t=129s

Recent Work in Embedded Visualization

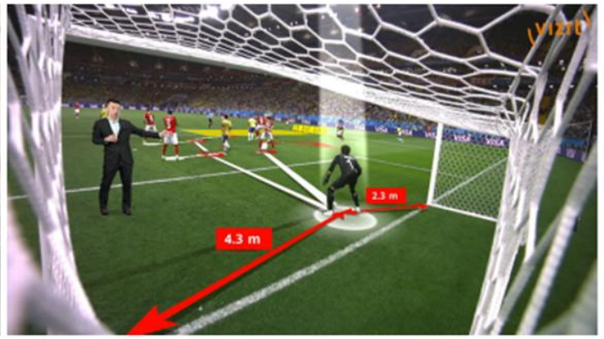
- **Embedded Vis** in sports can enhance viewer engagement without repeatedly **switching** between the digital and real worlds.



[VisCommentator, Chen, 2022]

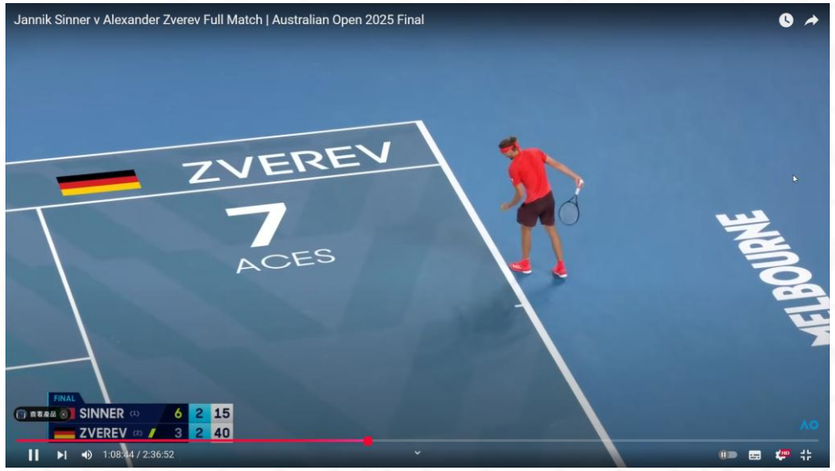


[SportsXR, SitVis for Free-throw, Lin, 2021]



[Viz Libero, Fischer, 2021]







Motivation

Embedded Visualization



[Omniocular, Lin, 2022]



[VisCourt, Cheng, 2024]

Coherence VR Tennis Narratives

- Enhance Comprehension
- Convey Narrative
- Ensure Visualization Clarity

Camera Motion



[Can You Move It, Zhu, 2023]



[GeoCamera, Li, 2023]



tracking to capture the player's elbow angle at ball strike.

Problem Space

- **RQ1:** How can the **cinematic language** of conventional tennis broadcasts be **adapted** to enhance narrative engagement **in VR**?

  ➡ **Contribution 1: Coding Annotations to Design Framework.**

- **RQ2:** How can dynamic camera movements and embedded visualizations be **integrated** to clearly communicate tactical and emotional match details in VR?

  ➡ **Contribution 2: Monocular-to-Immersive System Pipeline.**

- **RQ3:** To **what extent** does this integrated approach enhance viewer comprehension and engagement without compromising immersion and comfort?

  ➡ **Contribution 3: Results findings from a controlled user study.**

Annotation Results for Coding

- **2D Broadcasts:** Frequent **close-ups** & **angular pans** for emotional/tactical emphasis, Rapid cuts between angles to drive engagement
- **VR:** **Medium** shots & slow **linear** moves to balance narrative clarity and comfort, Predictable, **stable** framing to minimize disorientation

VR SHOT SIZE AND MOTION			*TENNIS BROADCAST AUDIO VISUAL LANGUAGE*							
Count	Tennis Scenario / Camera Motion	Primary Target / Shot Size	Action		Emotion			Tactic		*Total Count*
			Player	Ball in Play	Player	Stadium	Audience	Ball in Play	Stats Overlay	
23	Static	Wide Shot	0	0	0	14	8	0	8	30
40	Linear Motion	Wide Shot	0	0	1	12	6	0	0	19
0	Angular Motion	Wide Shot	0	0	0	26	6	7	1	40
94	Static	Medium Shot	6	3	7	20	11	1	8	56
124	Linear Motion	Medium Shot	12	18	9	11	9	23	0	82
13	Angular Motion	Medium Shot	1	4	8	13	6	72	0	104
61	Static	Close-Up	7	17	4	1	76	3	40	148
35	Linear Motion	Close-Up	9	23	17	1	13	1	0	64
13	Angular Motion	Close-Up	12	11	3	1	26	2	9	64
		Total Count	47	76	49	99	161	109	66	

Design Framework

Convey Narrative & Engagement

Shot Size & Angle

Wide | Medium | Close-Up


Birds' Eye


Baseline



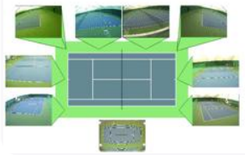
Court-Level


Stadium Corner



Sideline


Net-Cam



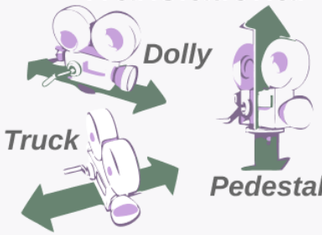

Corner


Follow-Cam

VR Camera Motion

Static | Translational | Rotational


Still





Enhance Comprehension

Event

Action | Emotion | Tactic

Embedded Vis

Metadata | Player | Ball



Panel

Label/Outline

Point

Floating Text

Line

Line

Volume

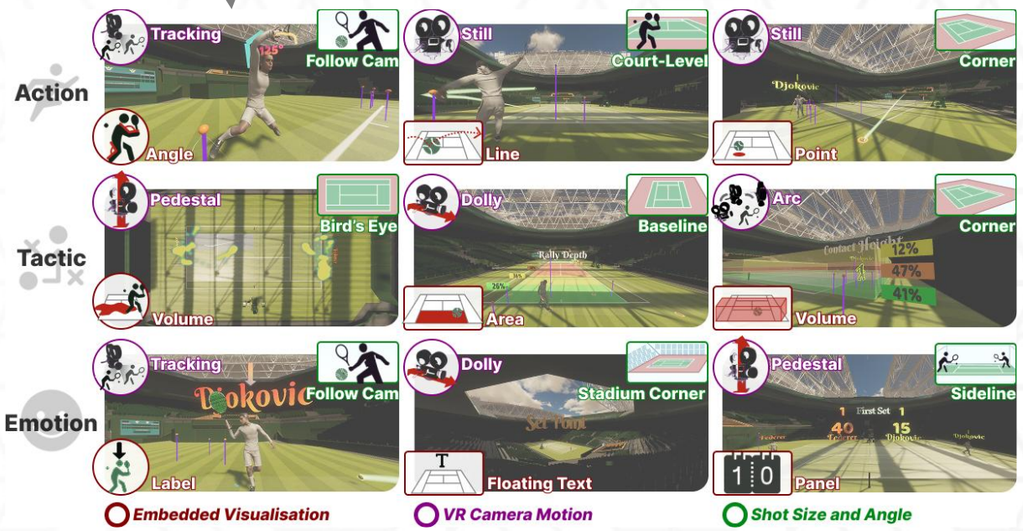
Area

Angle

Volume

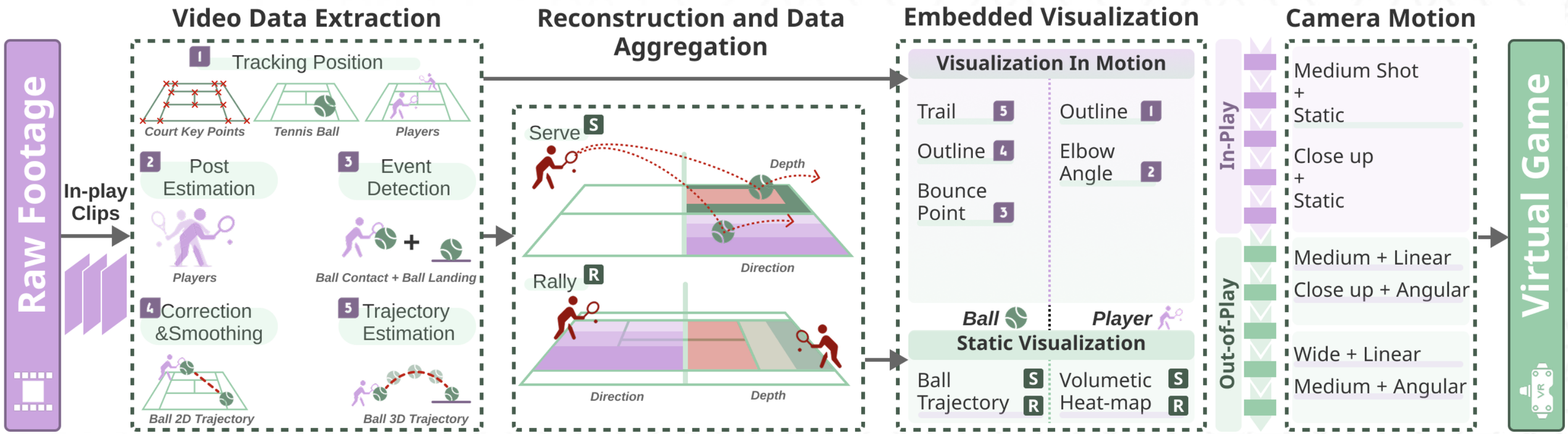
Ensure Visualization Clarity

Integrated Narrative Scenarios

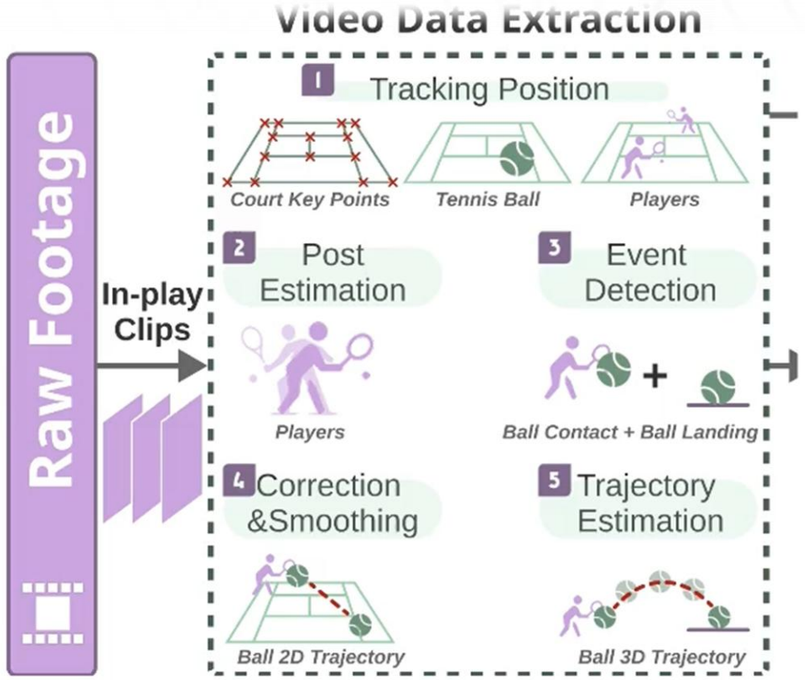


System Pipeline

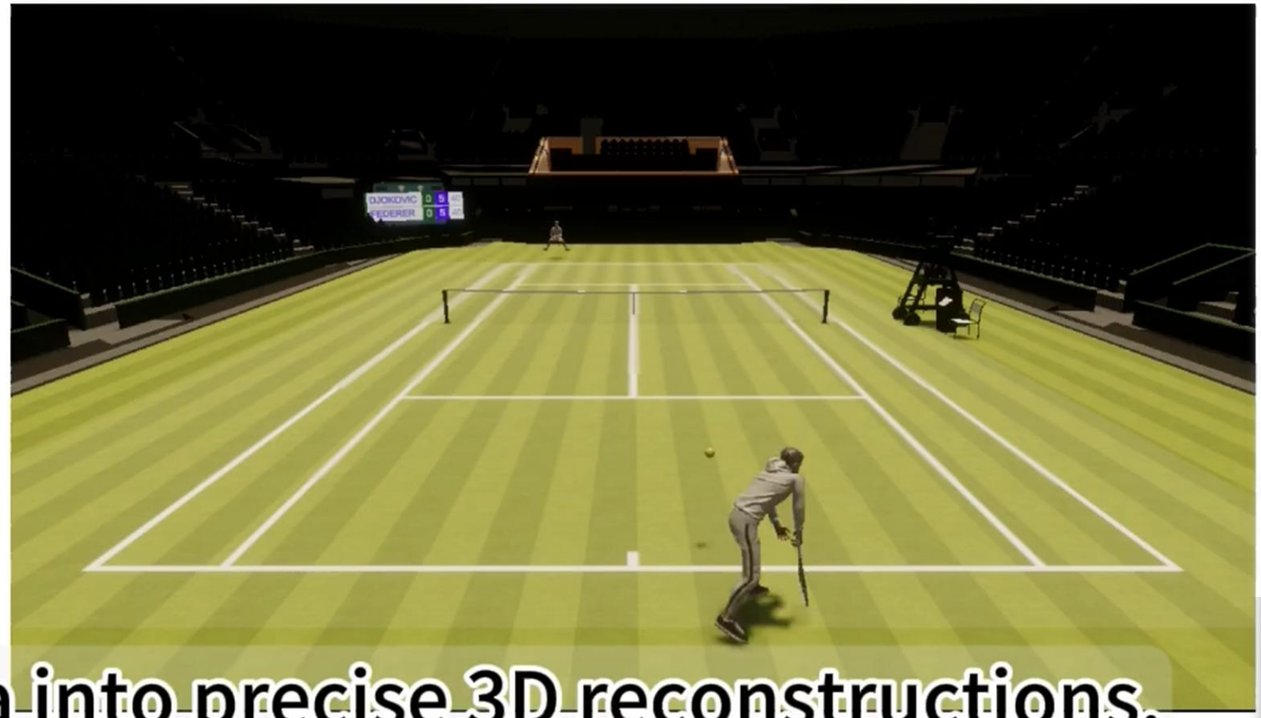
- **Semi**-automatic Video Data Extraction
- Real-time **3D Scene Rendering**
- Embedded Visualization & Camera Motion **Planning**



Data Extraction



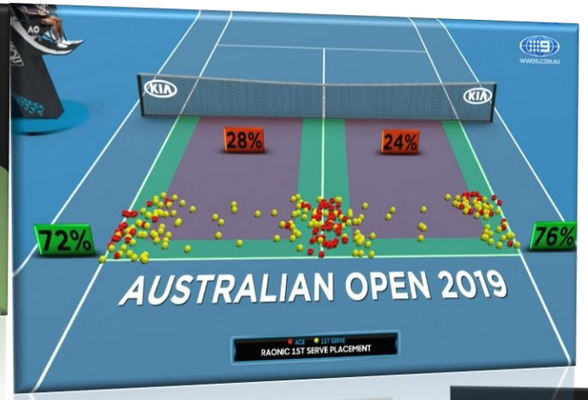


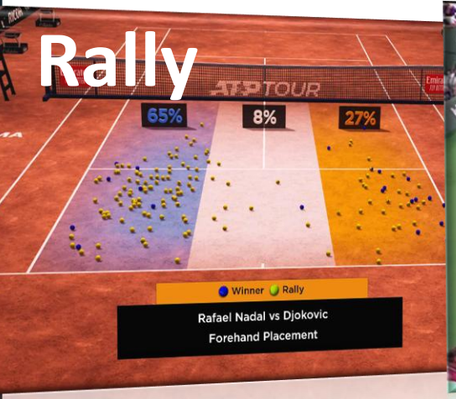


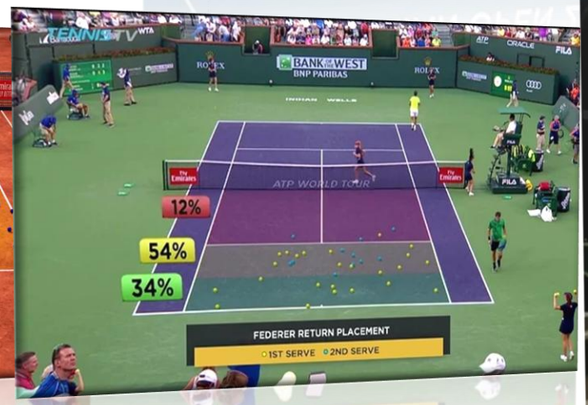
while corrections convert 2d data into precise 3D reconstructions.

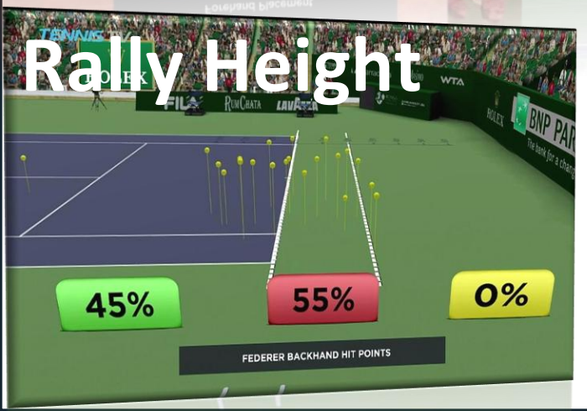
Reconstruction & Data Aggregation

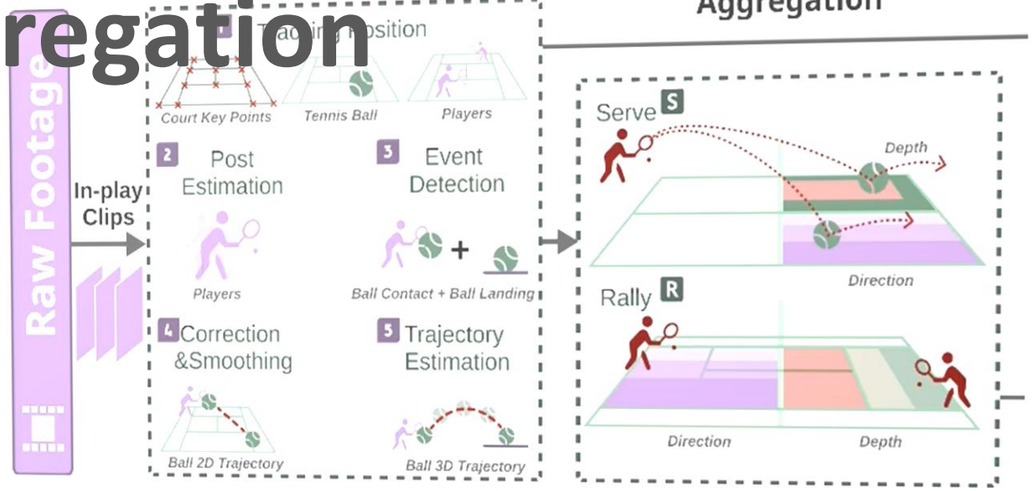


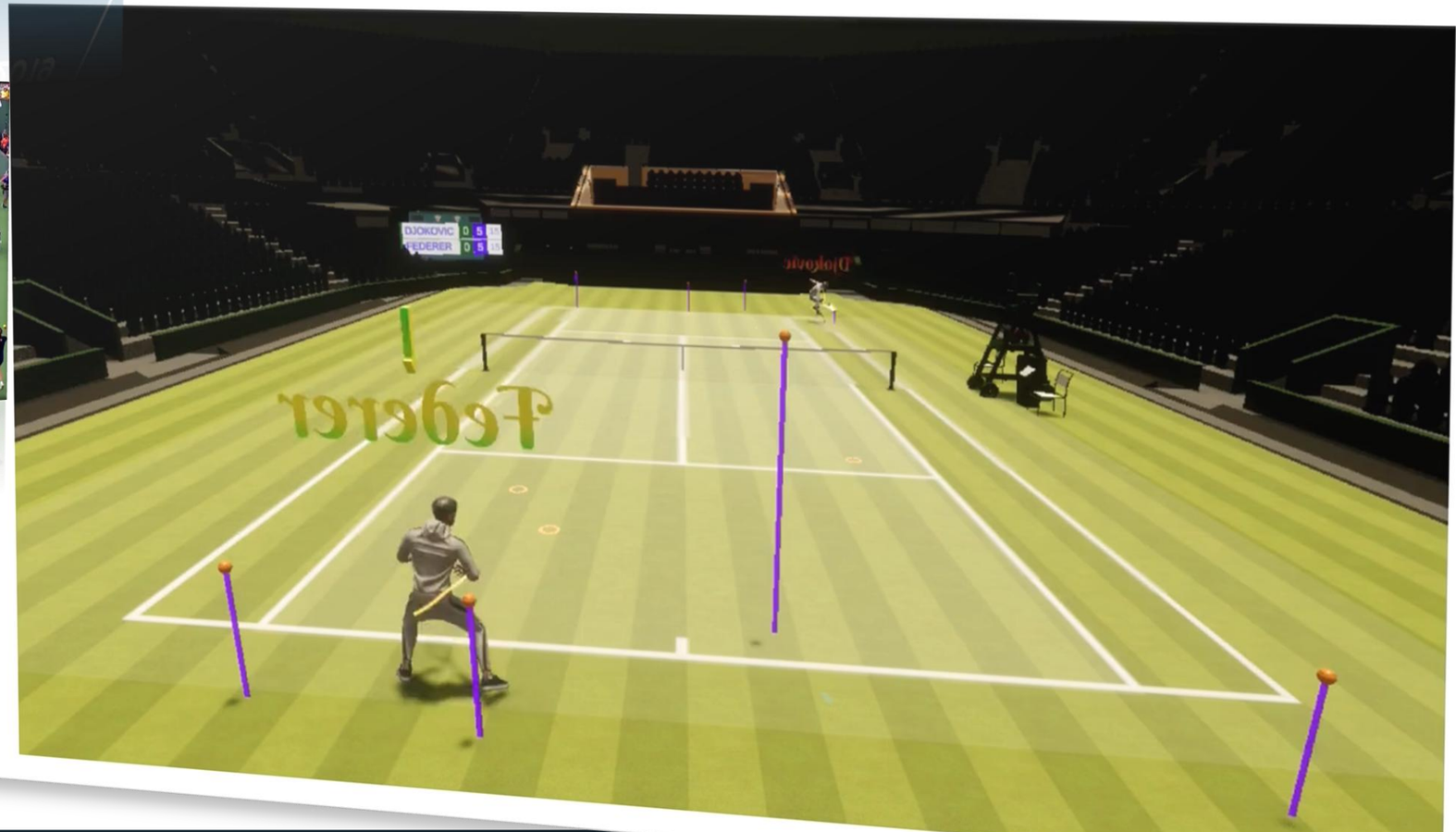












Within-subject User Study Design - 3 Conditions



User Study Results

16 Users

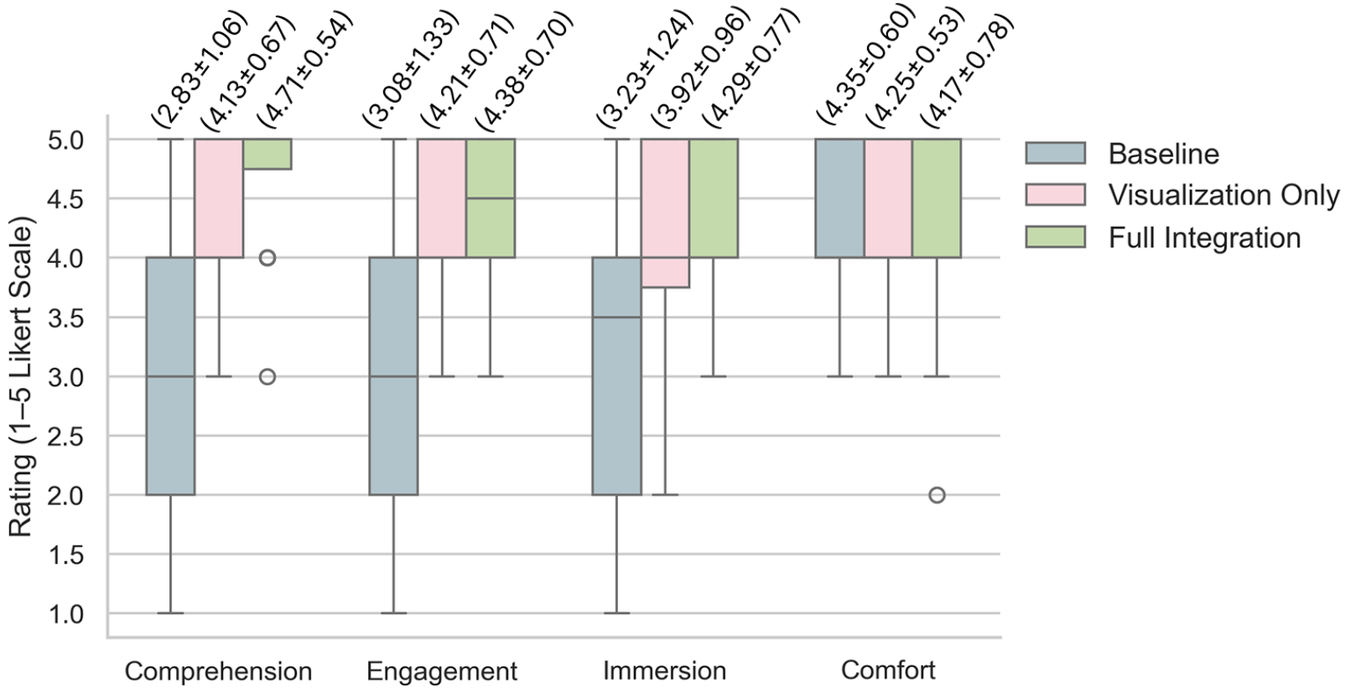
(Graduate students, diverse VR/tennis experience)

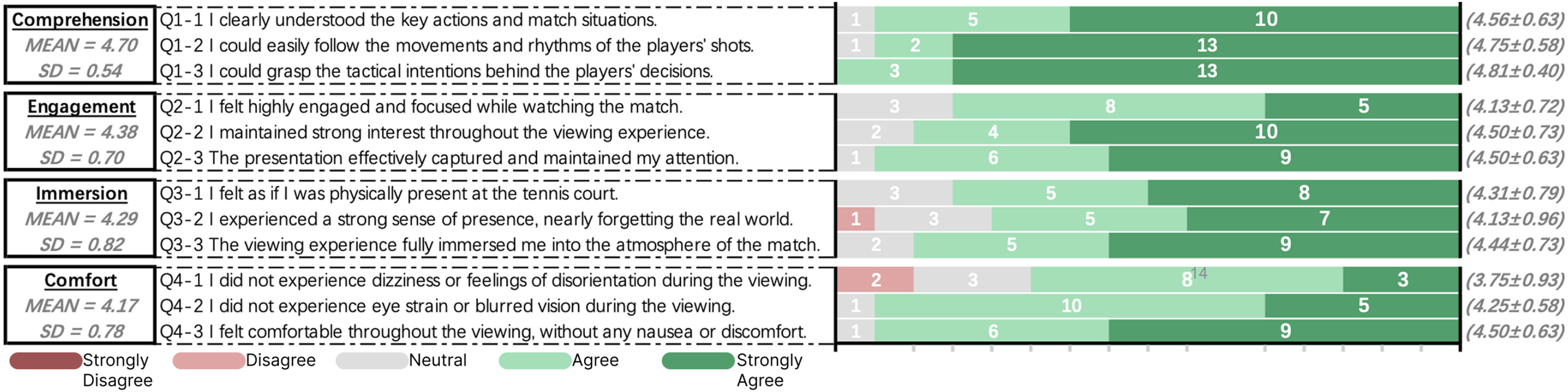
3 Conditions

(Baseline / Visualization Only / Full Integration)

4 Dimensions

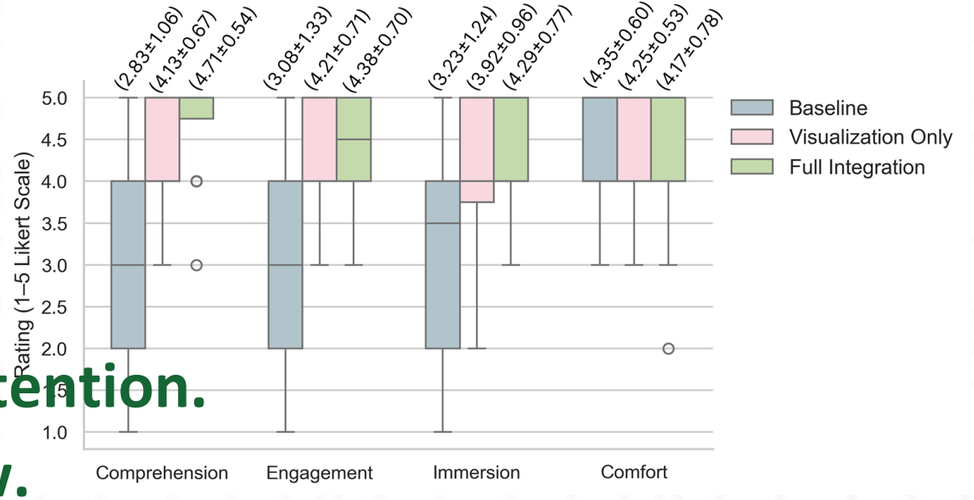
(Comprehension, Engagement, Immersion, Comfort)





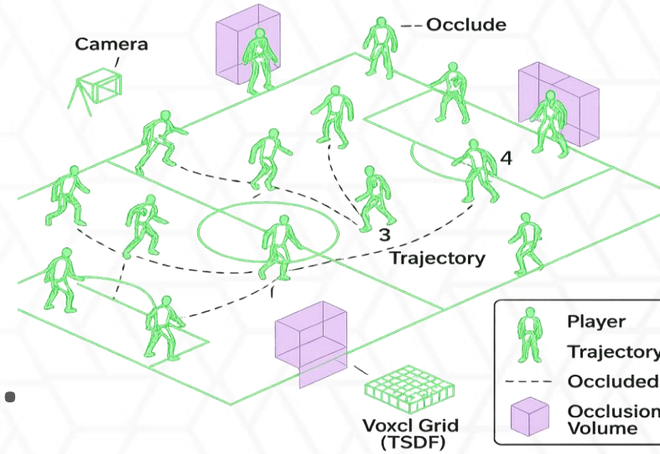
Findings

- **Comprehension ↑**
  - Embedded Vis + Camera Motion → **Guide Attention.**
  - Cognitive load ↓: Key events **easier to follow.**
- **Engagement/Immersion ↑**
  - **Dynamic framing** supports storytelling.
  - **Social presence lacking** → add spatial audio / commentary.
- **Comfort ↔ high**
  - High with **slow linear & gentle rotations**
  - Avoid vertical drops / sudden overhead cuts.
  - For **long rallies**: Baseline views to reduce head motion.
  - **Balance** overlay intensity & motion frequency.

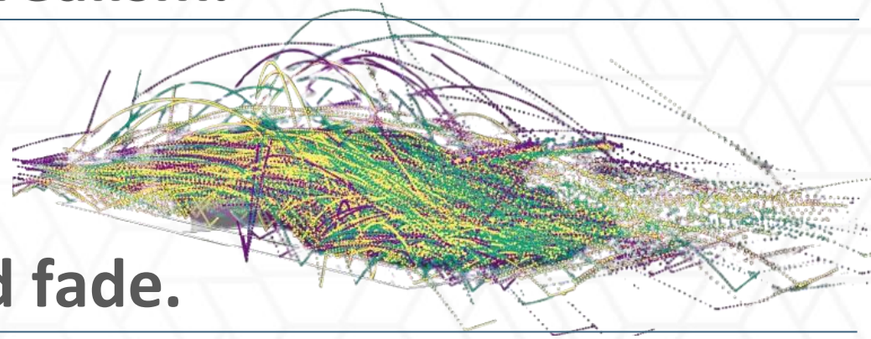


Takeaways

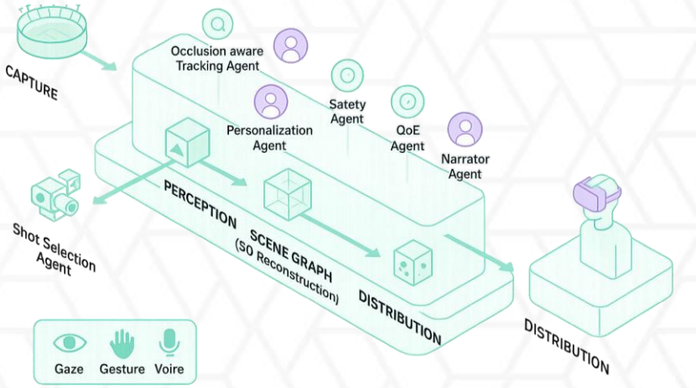
- **Broad Applicability**
  - Generalizes to **racket** sports.
  - **Team** sports → multi-agent + occlusion-aware tracking.
  - Current height estimates ≈ → **4D Gaussians** for realism.



- **Visualization Trade-offs**
  - **Dynamic** cues near action guide attention.
  - Cumulative trails **clutter** long rallies → pace and fade.



- **Balanced Viewer Agency**
  - **Passive** VR viewing can feel **isolating**.
  - Integrate **lightweight** controls
  - Using smart **production agents** to **personalize**.



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Thank you!

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